

# CariSurg MedTech Pathways Research Project

## Preliminary Proposal

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**Project Title:** Explainable AI Triage Assistant for Caribbean Emergency Departments

**Student Name:** Sariana Ramoutar

### Statement of the Problem:

Emergency departments rely on rapid triage decisions to prioritise patient care, yet heavy workloads, patient surges and human subjectivity can contribute to inconsistent triage outcomes. Recent studies have shown that artificial intelligence can support emergency department triage by improving speed and predictive performance, but many systems operate as “black boxes,” providing little explanation for how recommendations are made (Da’Costa et al., 2025; Tyler et al., 2025). This lack of explainability can make AI recommendations difficult for clinicians to understand, verify and confidently incorporate into their decision-making processes, limiting their practical use in healthcare settings (Bhuvaneswari et al., 2025).

### Previous Work:

Recent clinical literature highlights both the potential benefits and implementation challenges of AI within emergency triage.

- Multiple studies report that machine learning models frequently can outperform traditional triage approaches and support patient prioritisation, admission prediction and resource planning (Tyler et al., 2025; Da’Costa et al., 2025; Almulihi et al., 2024). Researchers have applied these models to predict immediate Emergency Severity Index (ESI) levels, downstream hospital admissions, ICU placements, and overall patient mortality risks (Agius et al., 2025; Almulihi et al., 2024; Tschoellitsch et al., 2023). Furthermore, advancements in Natural Language Processing (NLP) show that AI can extract clinically relevant information from free-text nursing notes that may be difficult for traditional systems to analyse (Lee et al., 2024).
- Triage accuracy is influenced by factors such as nurse experience and the amount of time available for patient assessment (Seo et al., 2024). Research suggests that clinicians often use “situational triage” to balance patient risks against operational pressures such as overcrowding and limited staffing. These findings support the development of human-centred AI frameworks that complement existing workflows (Gorick et al., 2023; Bhuvaneswari et al., 2025).
- Despite promising performance in research settings, several challenges remain before widespread clinical adoption. First, some studies report a tendency toward over-triage, which may increase pressure on limited healthcare resources (Tyler et al., 2025). Second, algorithmic bias remains a concern, particularly when models are trained on datasets that do not adequately represent diverse patient populations (Da’Costa et al., 2025; Huang et al., 2022). Finally, many published systems have limited external validation and are trained on isolated, single-institution datasets, making it difficult to determine how well they would perform in different healthcare settings (Agius et al., 2025; Araouchi & Adda, 2024; Tschoellitsch et al., 2023).

Collectively, these studies indicate that while strong predictive performance is achievable, a gap remains for transparent, explainable triage tools that align with existing clinical workflows.

**Proposed Solution:**

This project will develop and evaluate an explainable AI decision-support prototype optimised for emergency triage using a de-identified, publicly available dataset. In contrast to traditional “black box” systems, the proposed model will simultaneously predict patient acuity levels and generate a localised, clear map of the specific clinical inputs and vital signs influencing its recommendation.

The technical architecture will aim to achieve triage classification performance comparable to previously reported benchmark studies (e.g., Araouchi & Adda, 2024), while introducing explicit visual explainability features to address the trust and transparency concerns highlighted in the literature (Da’Costa et al., 2025). This prototype serves as a foundational step toward engineering equitable, interpretable triage software tailored specifically to the unique constraints, infrastructure and diverse patient demographics of Caribbean healthcare environments.

**References:**

- Agius, S., Cassar, V., Magri, C., Khan, W., Obe, D. A.-J., Caruana, G., & Topham, L. (2025). Predicting Emergency Severity Index (ESI) level, hospital admission, and admitting ward in an emergency department using data-driven machine learning. *BMC Medical Informatics and Decision Making*, 25(1), 281. <https://doi.org/10.1186/s12911-025-02941-9>
- Almulihi, Q., Alquraini, A., Almulihi, F., Alzahid, A., Qahtani, S., Almulhim, M., Alqhtani, S., Alnafeea, F., Mushni, S., Alaqil, N., Assiri, M., & Maghraby, N. (2024). Applications of Artificial Intelligence and Machine Learning in Emergency Medicine Triage—A Systematic Review. *Medical Archives*, 78(3), 198. <https://doi.org/10.5455/msm.2024.236-239>
- Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. *Procedia Computer Science*, 251, 430–437. <https://doi.org/10.1016/j.procs.2024.11.130>
- Bhuvaneswari, E., Prasad, K. D. V., Ashraf, M., Jadhav, S., Rao, T. R. K., & Sudha Rani, T. (2025). A human-centered hybrid AI framework for optimizing emergency triage in resource-constrained settings. *Intelligence-Based Medicine*, 12, 100311. <https://doi.org/10.1016/j.ibmed.2025.100311>
- Da’Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*, 197, 105838. <https://doi.org/10.1016/j.ijmedinf.2025.105838>
- Gorick, H., McGee, M., Wilson, G., Williams, E., Patel, J., Zonato, A., Ayodele, W., Shams, S., Di Battista, L., & Smith, T. O. (2023). Understanding triage assessment of acuity by emergency nurses at initial

- adult patient presentation: A qualitative systematic review. *International Emergency Nursing*, 71, 101334. <https://doi.org/10.1016/j.ienj.2023.101334>
- Huang, J., Galal, G., Etemadi, M., & Vaidyanathan, M. (2022). Evaluation and Mitigation of Racial Bias in Clinical Machine Learning Models: Scoping Review. *JMIR Medical Informatics*, 10(5), e36388. <https://doi.org/10.2196/36388>
- Lee, S., Lee, J., Park, J., Park, J., Kim, D., Lee, J., & Oh, J. (2024). Deep learning-based natural language processing for detecting medical symptoms and histories in emergency patient triage. *The American Journal of Emergency Medicine*, 77, 29–38. <https://doi.org/10.1016/j.ajem.2023.11.063>
- Seo, Y. H., Lee, K., & Jang, K. (2024). Factors influencing the classification accuracy of triage nurses in emergency department: Analysis of triage nurses' characteristics. *BMC Nursing*, 23(1), 764. <https://doi.org/10.1186/s12912-024-02334-9>
- Tschoellitsch, T., Seidl, P., Böck, C., Maletzky, A., Moser, P., Thumfart, S., Giretzlehner, M., Hochreiter, S., & Meier, J. (2023). Using emergency department triage for machine learning-based admission and mortality prediction. *European Journal of Emergency Medicine*, 30(6), 408–416. <https://doi.org/10.1097/MEJ.0000000000001068>
- Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Won Lee, D., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2025, September 24). *Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review*. Dr. Kiran C. Patel College of Osteopathic Medicine (KPCOM). Cureus. <https://doi.org/https://doi.org/10.7759/cureus.59906>

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